

Cognitive Deference to Automated Decision Aids: A Mixed-Methods Study of Trust Calibration Under Time Pressure

Thesis — Master of Arts

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Abstract

As automated decision aids enter high-tempo workplaces, a key question is whether people calibrate their trust to the aid's actual reliability or simply defer. This thesis combines a controlled experiment ($n = 96$) with follow-up interviews ($n = 18$) to examine trust calibration when a decision aid of known, imperfect accuracy is used under low versus high time pressure. Under time pressure, participants over-relied on the aid, accepting its recommendation even when it conflicted with strong contradicting evidence; interviews revealed that perceived authority and effort-saving, not assessed accuracy, drove deference. Results inform the design of aids that prompt deliberate, calibrated use.

Chapter 1: Introduction

Automated aids promise faster, more consistent decisions, but their value depends on users trusting them appropriately, neither ignoring a reliable aid nor deferring to an unreliable one. The phenomenon of over-reliance, where people accept an aid's output without scrutiny, is especially concerning under the time pressure typical of real work.

This thesis asks whether time pressure degrades trust calibration and what reasoning underlies deference. A mixed-methods design pairs an experiment that manipulates pressure with interviews that surface the why behind the behavior.

Chapter 2: Literature Review

Research on automation bias documents both omission and commission errors when people use decision aids, and trust-calibration theory frames appropriate reliance as a match between trust and demonstrated reliability. Less is known about how acute time pressure shifts this calibration, and quantitative studies rarely capture participants' own accounts.

This study addresses the gap by combining a controlled manipulation with qualitative interviews, allowing behavioral effects to be interpreted through participants' reasoning.

Chapter 3: Methods

In the experimental phase, 96 participants completed a judgment task with access to a decision aid described as accurate roughly 75 percent of the time, under randomly assigned low or high time pressure. The key measure was reliance on the aid when it conflicted with strong contradicting evidence.

In the qualitative phase, 18 participants completed semi-structured interviews analyzed with reflexive thematic analysis. The design, consent procedures, and analytic audit trail are described to support trustworthiness and replication.

Chapter 4: Results

Participants under high time pressure were significantly more likely to accept the aid's recommendation when it conflicted with strong evidence, indicating degraded calibration rather than uniformly higher trust.

Three interview themes emerged: deference to perceived authority of the system, relief at offloading effort, and a tendency to rationalize the aid's output after the fact. Assessed accuracy was rarely cited as a

reason for reliance.

Chapter 5: Discussion

The convergence of experimental and interview findings suggests that under pressure people substitute the aid's authority for their own judgment, a pattern consistent with cognitive deference rather than reasoned trust. This has direct design implications: aids should surface uncertainty and prompt verification rather than present confident single answers.

Limitations include a laboratory task that may understate real stakes, a student-weighted sample, and the modest qualitative sample. Threats to validity and reflexivity in the qualitative analysis are addressed.

Chapter 6: Conclusion

This thesis showed that time pressure undermines trust calibration in the use of automated decision aids and that deference is driven by perceived authority and effort-saving rather than assessed accuracy. The contribution is a mixed-methods account that links a behavioral effect to its underlying reasoning. Future work should test calibration-supporting interfaces in higher-stakes field settings.

References

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